

6. REST API: ORDER FROM THE MENU, NOT THE KITCHEN

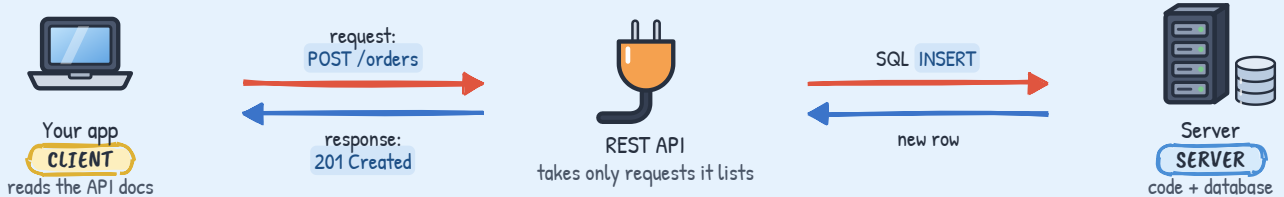
ANALOGY: A RESTAURANT

1 GUEST ORDERS, KITCHEN COOKS

One soup, please.



2 THE SAME IN SOFTWARE



The API is the menu: the requests a server accepts. Clients order from it, never walk into the kitchen.

THE THING AND THE VERB

- URL** • Names the thing: `/orders` is the list, `/orders/42` is one order.
- GET** • Read. `GET /orders/42` asks what is in order 42. Nothing changes.
- POST** • Create. `POST /orders` with a body adds a new order. The server picks its number.
- PUT** • Replace. `PUT /orders/42` sends the whole order as it should look now.
- PATCH** • Change a part. `PATCH /orders/42` with `{"dish": "salad"}` swaps only the dish.
- DELETE** • Remove. `DELETE /orders/42` cancels the order.

HOUSE RULES

- SAFE**
GET only looks. Asking what you ordered never changes the order.
- IDEMPOTENT**
Twice ends like once. GET, PUT and DELETE are; POST is not: two POSTs, two soups.
- STATELESS**
The waiter forgets you. Each request carries all it needs, even your token.

STATUS: 2XX WORKED, 4XX CLIENT, 5XX SERVER

200 OK Here is your dish: the data you asked for.	201 Created New order taken, usually after POST.	204 No Content Done, nothing to bring back. Common after DELETE.	400 Bad Request The order makes no sense: broken JSON.	401 Unauthorized Despite the name: we do not know who you are.
403 Forbidden We know you, but you may not do this.	404 Not Found Not on the menu: no such order.	405 Method Not Allowed This verb is not on the menu for this URL.	429 Too Many Requests Too many orders too fast. Wait, then retry.	500 Internal Server Error The kitchen broke. Not your fault.

ORDER FROM A TERMINAL (SAMPLE)

```
$ curl -i -X POST https://api.example.com/orders \
-H "Content-Type: application/json" \
-d '{"dish": "soup"}'
HTTP/2 201 ...headers...
{"id":42,"dish":"soup"}
api.example.com is a placeholder. -i shows the status line; without it you see only the body.
```



REMEMBER: Order from the menu: the URL names the thing, the method says what to do, and the status code tells you how it went.